

Scotia eCom+™

Hosted payment user guide.



Hosted payment user guide.

Hosted payment is the most agile and simple solution to implement since it allows you to include the acceptance of payments in a website with little development.

It consists of inserting an HTML form with the parameters of the transaction to be carried out and sending it to our Gateway. The payment process is completely managed by Scotiabank , so you do not need to build the payment gateway yourself and it is not necessary for the development team to have a PCI certification, since the tool is fully certified.

You can embed the payment gateway in an IFRAME within your site or redirect the customer in the browser to process the transaction. If you choose to integrate with IFRAME the client will never leave your website .

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Form customization

It is also worth mentioning that the payment forms are customizable in the productive environment through the Virtual Terminal tool, with which we can modify colors and add logos so that they adapt more to the style of our business.

Virtual Terminal->Customization->Customize the design of your hosted payment forms->Combined page

Design adjustments for hosted payment forms

Updated successfully

Classic Combined page

▼ Logo

► Background

Top Bar Body Content

Borders

► Text

Primary Title

► Buttons

Primary Hover

Version: 25.2.5.27 Build Date: 2023-05-11 11:07:22+0200

YOUR LOGO HERE

JoskuaStore

Please select payment method

Payment method

CANCEL CONTINUE

Your Order

Amount € 4.11 EUR

🔒

This is a secure page which uses SSL/TLS (Secure Socket Layer/Transport Layer Security) to encrypt and securely transmit your payment data.

YOUR LOGO HERE

What I need?

1. Have control over the source code of the frontend to insert the HTML form and backend to be able to correctly generate our HASH
2. **StoreID**. It is the identifier of your store within our platform.
3. **SharedSecret** It is a private key that should never be shared and should be stored securely. It is used to generate a HASH corresponding to each transaction and to guarantee its authenticity at the Gateway.
4. Valid SSL certificate for security reasons.
5. Own URL to receive the response (success or error) to the transactions in the form of POST parameters .

Integration process

Done, you have decided that Connect is the solution for your business! Here are the steps to follow:

Get in touch with a commercial advisor to carry out your affiliation process.

Via email you will receive your **SharedSecret**, **StoreID** and test card numbers so you can start your development.

Once your integration is ready in the test environment, give notice to coordinate a certification session , in this session we will ask you to make some test sales and we will test success and rejection cases where we will ask you to:

The correct amount is displayed to the cardholder.
The oid associated with the transaction is shown, this is for support issues since in this way for any question it will be much easier to find the transaction in our system, and provide you with a better service.

In case of rejection a correct handling of the error.
When your business has been successfully certified then your productive credentials will be released with a new SharedSecret and StoreID.

Connect Integration Guide

Here are the steps for development.

Hash Generation

Each of the transactions carried out through IPG's Connect solution requires the inclusion of a security hash in order to guarantee the integrity of the message on our server.

For this we will need at least the following parameters within our HTML form in the following order (it is possible that depending on the operation more parameters are required)

Connect Integration Guide

Field Name	Description	Example
chargetotal	This is the total amount of the transaction using a period or a comma as the decimal separator	13
checkoutoption	Set the value of this parameter to "combinedpage" for a standard checkout page integration.	combinedpage
currency	The numeric ISO code of the currency of the transaction. 484 is the code for Mexican pesos, 840 USD.	484
hash_algorithm	This is to indicate the algorithm it uses for hash calculation. Possible values is: HMACSHA256	HMACSHA256
responseFailURL	The URL you want to direct customers to after a declined or failed transaction	https://localhost:8643/webshop/response_failure.jsp
responseSuccessURL	The URL you want to direct customers to after an accepted or successful transaction	https://localhost:8643/webshop/response_success.jsp
storename	It is the StoreId that we will provide you by mail, it always starts with 62	62666666
timezone	Transaction time zone in Area/Location format	America/Mexico_City
txndatetime	YYYY:MM:DD-hh:mm:ss exact time of the transaction. A fixed value should not be used.	2022:05:11-12:32:45
txntype	'sale', 'preauth', 'postauth', or 'payer_auth'	sale
sharedsecret	This is the key used that we have shared to calculate the hash	TopSecret

Connect Integration Guide

Step 1 - Extended Hash needs to be calculated using **ALL HTML form parameters ascending by field name** (except sharedsecret and hashExtended), where uppercase parameters must come before lowercase ones (Based on ASCII values).

Later join the parameters separated by pipes. **Only use the value of the parameters and not their name.**

The hash string does not include the 'sharedsecret' key, this is used as the secret key or encryption key for the HMAC algorithm.

```
stringToExtendedHash =  
13.00|combinedpage|978|HMACSHA256|https://localhost:8643/webshop/response_failure.jsp|http  
s:  
//localhost:8643/webshop/response_success.jsp|10123456789|Europe/Berlin|2021:09:06-  
16:43:04|sale
```

Step 2 - Pass the previously created string through the HMCASHA256 algorithm and use the Secret that was shared as the key to generate the calculated Hash value.

```
HmacSHA256(stringToExtendedHash, sharedsecret)
```

Step 3 - Encode the return value of the HMACSHA256 algorithm using Base64 and pass it as part of the HTML form.

```
Base64:  
EapafBqqOF6N/kch8USkHPGh+fwSko24h6FpQnQHfQ8=  
<input type="hidden" name="hashExtended"  
value="EapafBqqOF6N/kch8USkHPGh+fwSko24h6FpQnQHfQ8="/>
```

Hash Validation

A very good security practice is also to validate the hash obtained from the response of each transaction, this to validate that the integrity of the message has been preserved when obtaining a response.

To do this, we will verify if the hash obtained in the response under the `response_hash` parameter matches the hash calculated with our Secret, by evaluating the following string with the values obtained in the response:

`approval_code|chargetotal|currency|txndatetime|storename`

Direct Sale

Below is a form that represents a sale of \$100.00 MXN that is being made on January 02, 2020 at 12:51:05 PM that is being made on the store with ID 399000002. At the end of the transaction, the buyer is redirected to the URLs specified in **responseFailURL** or **responseSuccessURL**.

```
<form method="POST" action="https://test.ipg-  
online.com/connect/gateway/processing">  
  <input type="hidden" name="chargetotal" value="100.00" />  
  <input type="hidden" name="checkoutoption" value="combinedpage" />  
  <input type="hidden" name="currency" value="484" />  
  <input type="hidden" name="hash_algorithm" value="HMACSHA256" />  
  <input type="hidden" name="hashExtended"  
value="EapafBqqOF6N/kch8USkHPGh+fwSko24h6FpQnQHfQ8=" />  
  <input type="hidden" name="responseFailURL"  
value="https://myurl.com/checkout/success" />  
  <input type="hidden" name="responseSuccessURL"  
value="https://myurl.com/checkout/error" />  
  <input type="hidden" name="storename" value="399000002" />  
  <input type="hidden" name="timezone" value="America/Mexico_City" />  
  <input type="hidden" name="txndatetime" value="2020:01:02-12:51:05" />  
  <input type="hidden" name="txntype" value="sale" />  
</form>
```

IFRAME

If you want to embed your form inside an iframe follow these steps

Add the following javascript code snippet to your "Checkout" page. In this way you can receive notification of the transactions made and redirect the buyer to the "SUCCESS" or "ERROR" page.

```
window.addEventListener( "message", receiveMessage, false) ;
```

```
function receiveMessage(event) {  
  if (event.origin "https://test.ipg-online.com")  
    return;  
  var elementArr = event.data.elementArr;  
  forwardForm(event.data, elementArr);  
}  
function forwardForm( responseobj, elementArr) {  
  var newForm = document.createElement ("form");  
  newForm.setAttribute('method', ' post ');  
  newForm.setAttribute('action ', responseObj.redirectURL);  
  newForm.setAttribute('id', 'newForm ');  
  newForm.setAttribute('name', 'newForm');  
  document.body.appendChild(newForm) ;  
  for (var i = 0; i < elementArr.length; i++) {  
    var element = elementArr[i];  
    var input = document.createElement(' input' );  
    input.setAttribute( 'type ', 'hidden ');  
    input.setAttribute( ' name ', element.name) ;  
    input.setAttribute( ' value ', element.value) ;  
    document.newForm.appendChild (input) ;  
  }  
  document.newForm.submit( ) ;  
}
```

IFRAME

Create a simple HTML payment form within your website and on the "Checkout" page. Where in addition to the Direct Sales parameters we will add the target attribute to the form element and the hostURI parameter

```
<form method="POST" target="myFrame" action="https://test.ipg-  
online.com/connect/gateway/processing">  
  <input type="hidden" name="chargetotal" value="100.00" />  
  <input type="hidden" name="checkoutoption" value="combinedpage" />  
  <input type="hidden" name="currency" value="484" />  
  <input type="hidden" name="hash_algorithm" value="HMACSHA256" />  
  <input type="hidden" name="hashExtended"  
value="EapafBqqOF6N/kch8USkHPGh+fwSko24h6FpQnQHfQ8=" />  
  <input type="hidden" name="hostURI" value="URL WHERE IFRAME IS  
LOADED" />  
  <input type="hidden" name="responseFailURL"  
value="https://myurl.com/checkout/success" />  
  <input type="hidden" name="responseSuccessURL"  
value="https://myurl.com/checkout/error" />  
  <input type="hidden" name="storename" value="399000002" />  
  <input type="hidden" name="timezone" value="America/Mexico_City" />  
  <input type="hidden" name="txndatetime" value="2020:01:02-12:51:05" />  
  <input type="hidden" name="txntype" value="sale" />  
</form>  
<iframe name="myFrame"></iframe>
```

Error handling - Connect

Issuer rejections

To correctly display the information on issuer rejections within the Connect solution, it is necessary to read the association Response Code parameter obtained from the response to the URL specified as POST parameters.

To interpret these errors we can rely on the following table. Soft errors must be distinguished from Hard errors.

Soft errors: they will allow us to retry the transaction.

Hard errors: no matter how many times we try the transaction, it will always be declined.

Error handling - Connect

Association Response Code	Message	Description	Soft vs Hard
00	Y:[AuthCode]:[Transaction ID]:[AVSResult]:[Endpoint Transaction ID]	Approved	
01	N:01:01-REFERRAL	The transaction was not authorised by the Issuer for an unknown reason. The cardholder should contact their Issuer.	Hard
02	N:01:02-REFERRAL	The transaction was not authorised by the Issuer for an unknown reason. The cardholder should contact their Issuer.	Hard
03	N:03:03-INVLD MER ID	The transaction was declined as it was invalid or not enabled for the merchant.	Hard
04	N:60:04-HOLD-CALL CTR	The transaction was not authorised by the Issuer most likely because fraud was suspected.	Hard
05	N:51:05-DECLINED	The transaction was not authorised by the Issuer for an unknown reason. (Do Not Honor)	Hard
06	N:01:06-DECLINED	An unknown error occurred when processing the transaction. Retrying the transaction may result in an Approval.	Soft
07	N:60:07-HOLD-CALL CTR	The transaction was not authorised by the Issuer most likely because fraud was suspected.	Hard
12	N:12:12-INV TRAN	The transaction was declined as it was invalid or not enabled for the merchant.	Hard
13	N:13:13-INV AMT The amount is set to zero, is unreadable, or exceeds the allowable amount.	Make sure the amount is greater than zero and in a suitable format.	Soft
14	N:14:14-INV ACCT NUM	The submitted Card number does not include the proper number of digits. The customer should try again using the correct card number.	Soft
15	N:51:15-DECLINED	An Issuer related to this card number has not been found.	Hard

Error handling - Connect

Association Response Code	Message	Description	Soft vs Hard
30	N:91:30-PLEASE RETRY	The transaction was declined, however, retrying the transaction may result in an Approval	Soft
33	N:54:33-EXPIRED CARD	The Card is expired. The customer will need to use a different payment method.	Hard
39	N:51:39-DECLINED	The transaction has been declined because no such credit card account exists.	Hard
41	N:60:41-HOLD-CALL CTR	The card was identified as lost by the Issuer and will not be processed.	Hard
43	N:60:43-HOLD-CALL CTR	The card was identified as stolen by the Issuer and will not be processed.	Hard
51	N:51:51-DECLINED	The cardholders account does not have sufficient funds to cover the transaction amount. Subsequent attempts at a later date may be successful.	Soft
54	N:54:54-EXPIRED CARD	The Card is expired. The customer will need to use a different payment method.	Hard
55	N:55:55-INCORRECT PIN	The PIN or Code entered is incorrect. Try again with another.	Soft
57	N:57:57-TXN NOT ALLOW	The transaction is not allowed by the Issuer.	Hard
58	N:63:58-SERV NOT ALLOWED	The terminal has not been configured to accept the transaction.	Hard
61	N:51:61-TRY AGAIN LATER	The cardholders account does not have sufficient funds to cover the transaction amount. Subsequent attempts at a later date may be successful.	Soft
62	N:63:62-SER NOT ALLOW	Approved	
65	N:51:65-TRY AGAIN LATER	The card has exceeded the frequency of transactions allowed.	Soft
68	N:91:68-PLEASE RETRY	The response was received too late and a timeout occurred. Retrying the transaction may result in an Approval.	Soft

Error handling - Connect

Association Response Code	Message	Description	Soft vs Hard
75	N:75:75-PIN RETRY MAX	The Issuer has declined the transaction because the cardholder has entered the Pin incorrectly too many times.	Soft
91	N:91:91-PLEASE RETRY	The Issuer is currently unavailable. This may be temporary - a subsequent attempt may be successful.	Soft
92	N:91:92-PLEASE RETRY	The Issuer is currently unavailable. This may be temporary - a subsequent attempt may be successful.	Soft
94	N:94:94-DUPLICATE TRAN	The submitted transaction appears to be a duplicate of a previously submitted transaction and was declined to prevent charging the same card twice for the same service. Check your systems to see if a duplicate exists.	Hard
96	N:CE:96-SYSTEM PROBLEM	There is a system error which has prevented the transaction from being processed. Retrying the transaction may result in an Approval.	Soft

Internal errors

Internal error handling for the Connect solution is done by evaluating the fail_rc parameter in conjunction with the table below.

Code	Description
5018	No transaction found for void
2303	Invalid credit card number
30051	Transaction timed out
5103	Cardholder did not return from ACS
2304	Invalid or expired card expiry date
5101	3D Secure authentication failed
5315	Transaction is already in state DECLINED
5005	FRAUD - Purchase limit exceeded
5002	brand not supported
5993	Cancelled by user
30031	No terminal setup
7778	Transaction timed out
50698	Incompatible referenced transaction
5003	The order already exists in the database.
5005	FRAUD - Duplicate lockout
5009	No transaction to return found
7777	System too busy
5996	Invalid card type
95001	Currency not supported for Local cards
5995	order too old to be referenced
50004	Invalid transaction request
96000	installment not supported

Internal errors

Internal error handling for the Connect solution is done by evaluating the fail_rc parameter in conjunction with the table below.

Code	Description
5994	The selected brand does not match the card number.
5102	ECI 7
50687	Split Authentication not supported
5022	not voidable
5014	Validation problem
10601	Total amount passed is more than the Return/Void amount.
10501	PostAuth already performed

Additional Features

In addition to Direct Sales, Connect offers the functionality of being able to process the following transactions. It is important to mention that if any of these services are required, mention should be made to the Fiserv Integrations Team so that the service is added at the time of store generation.

MSI Sale

To make a monthly sale without interest, it is practically the same as a Direct Sale with the difference that we will add two additional parameters to our HTML form which must be considered when generating the hash.

The form below represents a sale of \$100.00 MXN that is being made on January 02, 2020 at 12:51:05 PM that is being made on the store with ID 399000002. Additionally, two parameters were added: `numberOfInstallments`, which represents the number of months to pay; and `installmentsInterest` indicating whether interest will be applied or not. In addition, it is possible to add the `installmentDelayMonths` parameter to make the first payment after the N months specified in it.

Additional Features

```
<form method="POST" action="https://test.ipg-
online.com/connect/gateway/processing">
    <input type="hidden" name="chargetotal" value="100.00" />
    <input type="hidden" name="checkoutoption"
value="combinedpage" />
    <input type="hidden" name="currency" value="484" />
    <input type="hidden" name="hash_algorithm"
value="HMACSHA256" />
    <input type="hidden" name="hashExtended"
value="EapafBqqOF6N/kch8USkHPGh+fwSko24h6FpQnQHfQ8=" />
    <input type="hidden" name="installmentsInterest" value="false" />
    <input type="hidden" name="numberOfInstallments" value="6" />
    <input type="hidden" name="responseFailURL "
value="https://myurl.com/checkout/success" />
    <input type="hidden" name="responseSuccessURL "
value="https://myurl.com/checkout/error" />
    <input type="hidden" name="storename" value="399000002" />
    <input type="hidden" name="timezone"
value="America/Mexico_City" />
    <input type="hidden" name="txndatetime" value="2020:01:02-
12:51:05" />
    <input type="hidden" name="txntype" value="sale" />
</form>
```

Tokenization

To make a sale and tokenize a card it is practically the same as a Direct Sale with the difference that we will add an additional parameter to our HTML form which must be considered when generating the hash.

Upon successful processing of the transaction, the generated token can be obtained in the response received as POST parameters to your URL specified in `responseSuccessURL` under the `hosteddataid` field . This token will be associated within our Gateway to the payment method used for the operation, it is your duty to implement the logic to save it within your database and associate it with your client.

```
<form method="POST" action="https://test.ipg-  
online.com/connect/gateway/processing">  
  
  <input type="hidden" name="assignToken" value="true" />  
  <input type="hidden" name="chargetotal" value="100.00" />  
  <input type="hidden" name="checkoutoption" value="combinedpage" />  
  <input type="hidden" name="currency" value="484" />  
  <input type="hidden" name="hash_algorithm" value="HMACSHA256" />  
  <input type="hidden" name="hashExtended"  
value="EapafBqqOF6N/kch8USkHPGh+fwSko24h6FpQnQHfQ8=" />  
  <input type="hidden" name="responseFailURL"  
value="https://myurl.com/checkout/success" />  
  <input type="hidden" name="responseSuccessURL"  
value="https://myurl.com/checkout/error" />  
  <input type="hidden" name="storename" value="399000002" />  
  <input type="hidden" name="timezone" value="America/Mexico_City" />  
  <input type="hidden" name="txndatetime" value="2020:01:02-12:51:05" />  
  <input type="hidden" name="txntype" value="sale" />  
</form>
```

Tokenization

Use of the token

When sending the `hosteddataid` parameter with a previously generated token, the buyer will only be asked to enter the Security Code (CVV/CVV2) used for that payment method; in this way the IPG Gateway will obtain the encrypted data of the card and will process the transaction.

```
<form method="POST" action="https://test.ipg-  
online.com/connect/gateway/processing">  
  <input type="hidden" name="chargetotal" value="100.00" />  
  <input type="hidden" name="checkoutoption" value="combinedpage" />  
  <input type="hidden" name="currency" value="484" />  
  <input type="hidden" name="hash_algorithm" value="HMACSHA256" />  
  <input type="hidden" name="hashExtended"  
value="EapafBqqOF6N/kch8USkHPGh+fwSko24h6FpQnQHfQ8=" />  
  <input type="hidden" name="hosteddataid" value="TOKEN_TO_USE" />  
  <input type="hidden" name="responseFailURL"  
value="https://myurl.com/checkout/success" />  
  <input type="hidden" name="responseSuccessURL"  
value="https://myurl.com/checkout/error" />  
  <input type="hidden" name="storename" value="399000002" />  
  <input type="hidden" name="timezone" value="America/Mexico_City" />  
  <input type="hidden" name="txndatetime" value="2020:01:02-12:51:05" />  
  <input type="hidden" name="txntype" value="sale" />  
</form>
```

Scheduled Recurring Charges

Through connect it is also possible to generate recurring charges that are repeated a finite number of times.

For this we will add additional parameters to our HTML form which must be considered when generating the hash.

The form below represents a sale of \$100.00 MXN that is being made on January 02, 2020, at 12:51:05 PM that is being made on the store with ID 399000002. This charge will be made automatically 12 times (recurringInstallmentCount), each charge will be made every 1 (recurringInstallmentFrequency) month (recurringInstallmentPeriod). It is MANDATORY to send a contract number for the transaction in the ponumber parameter .

```
<form method="POST" action="https://test.ipg-online.com/connect/gateway/processing">
  <input type="hidden" name="chargetotal" value="100.00" />
  <input type="hidden" name="checkoutoption" value="combinedpage" />
  <input type="hidden" name="currency" value="484" />
  <input type="hidden" name="hash_algorithm" value="HMACSHA256" />
  <input type="hidden" name="hashExtended"
value="EapafBqqOF6N/kch8USkHPGh+fwSko24h6FpQnQHfQ8=" />
  <input type="hidden" name="ponumber" value="PO02220202" />
  <input type="hidden" name="recurringInstallmentCount" value="12" />
  <input type="hidden" name="recurringInstallmentPeriod" value="month" />
  <input type="hidden" name="recurringInstallmentFrequency" value="1" />
  <input type="hidden" name="recurringComments" value="Comentarios adicionales de la
transacción" />
  <input type="hidden" name="responseFailURL" value="https://myurl.com/checkout/success"
/>
  <input type="hidden" name="responseSuccessURL"
value="https://myurl.com/checkout/error" />
  <input type="hidden" name="storename" value="399000002" />
  <input type="hidden" name="timezone" value="America/Mexico_City" />
  <input type="hidden" name="txndatetime" value="2020:01:02-12:51:05" />
  <input type="hidden" name="txntype" value="sale" />
</form>
```


Thank you.